MUDFLOW OS | DAILY DRILLING FLUID REPORT

REPORT NO: #14
DATE: 2026-10-14

Operator: Equinor Energy | **Rig:** Deepsea Nordkapp | **Well:** Castberg 7220/8-1 | **Field:** Johan Castberg

Status: Active Drilling

1. OPERATIONAL & DRILLING PARAMETERS

Measured Depth (MD):	11,250.0 ft	**Mud Weight:**	12.40 ppg	**Flow Rate:**	520 GPM
True Vertical Depth:	9,820.0 ft	**Mud Type:**	Synthetic OBM (Versaclean)	**Pump Pressure:**	3150 psi
Daily Footage:	420.0 ft	**Hole Size / ID:**	8.500 in	**Bit TFA / Nozzles:**	0.451 in² (3x14)
Current Operation:	Drilling 8-1/2 section through Stø sandstone	**Formation:**	Stø Formation	**Drill Pipe OD:**	5.00 in

2. VISCOMETER READINGS & RHEOLOGICAL PROFILE (FANN 35 / API RP 13D)

600 RPM	300 RPM	200 RPM	100 RPM	6 RPM	3 RPM	10s Gel	10m Gel	30m Gel
63.0	42.0	32.0	20.0	7.0	5.0	6.0	11.0	14.0

tic Viscosity (PV):	21.0 cP	**Yield Point (YP):**	15.0 lb/100ft²	**Apparent Viscosity (AV):**	31.0 cP	**YP / PV Ratio:**	0.71
er Law (n):	0.5840	**Power Law (K):**	1.1240	**Herschel-Bulkley (Tau0):**	4.26 lb/100ft²	**Rheology Model:**	Hersche kley (AP 13D)

3. HYDRAULICS, ECD & BIT PERFORMANCE PROFILE

Hydrostatic Pressure:	6,332.0 psi	**Annular Pressure Loss:**	145.0 psi	**ECD at Total Depth:**	**12.68 ppg**
Jet Nozzle Velocity:	370.1 ft/sec	**Bit Pressure Drop:**	1,328.0 psi	**Bit Hydraulic Power:**	402.9 HHP
HSI (HHP / sq.in):	7.10 hp/in²	**Jet Impact Force:**	1,240.2 lbf	**Annular Velocity (DP):**	134.2 ft/min
Annular Flow Regime:	Laminar Flow (Optimal Wellbore Stability)	**Cuttings Transport:**	Optimal Laminar Cleaning	**Bit Pressure %:**	42.2% of pump

4. RETORT, SOLIDS CONTROL & VOLUMETRIC ACCOUNTING

Retort Water:	14.5 % vol	**Active Pits:**	680.0 bbl
Retort Oil:	66.5 % vol	**Reserve Pits:**	450.0 bbl
Retort Solids:	19.0 % vol	**Trip Tank:**	42.0 bbl
Average Solids SG (ASG):	3.82	**Total In-Hole:**	512.0 bbl
High Gravity Solids (HGS):	14.49 % vol	**Total System Volume:**	**1,684.0 bbl**
Low Gravity Solids (LGS):	4.51 % vol	**Chlorides:**	18,500 mg/L
Active Clay (MBT):	12.0 ppb eq	**LGS Target Status:**	Within Target (< 5.5%)

5. CHEMICAL ADDITIONS & INVENTORY

Product / Additive	Packaging / Unit	Daily Addition	Previous Stock	Ending Stock
Barite (API 4.2 SG)	100 lb Sack	**120**	1400	1280

EDC 99-DW Base Oil	BBL	**35**	400	365
Invermul Primary Emulsifier	55 Gal Drum	**2**	24	22
EZ-MUL NT Secondary Emulsifier	55 Gal Drum	**2**	20	18
Geltone V Organophilic Clay	50 lb Sack	**15**	150	135

6. ENGINEER REMARKS & OPERATIONAL RECOMMENDATIONS

Properties within target specifications. Good shaker performance on 170 mesh. Maintained ECD below fracture gradient window (13.1 ppg).

Lead Mud Engineer: Erik Lindqvist (Senior Drilling Fluids Specialist)

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Company Drilling Rep (Superintendent): _____

Signature / Approval: [VERIFIED ELECTRONIC LOG]